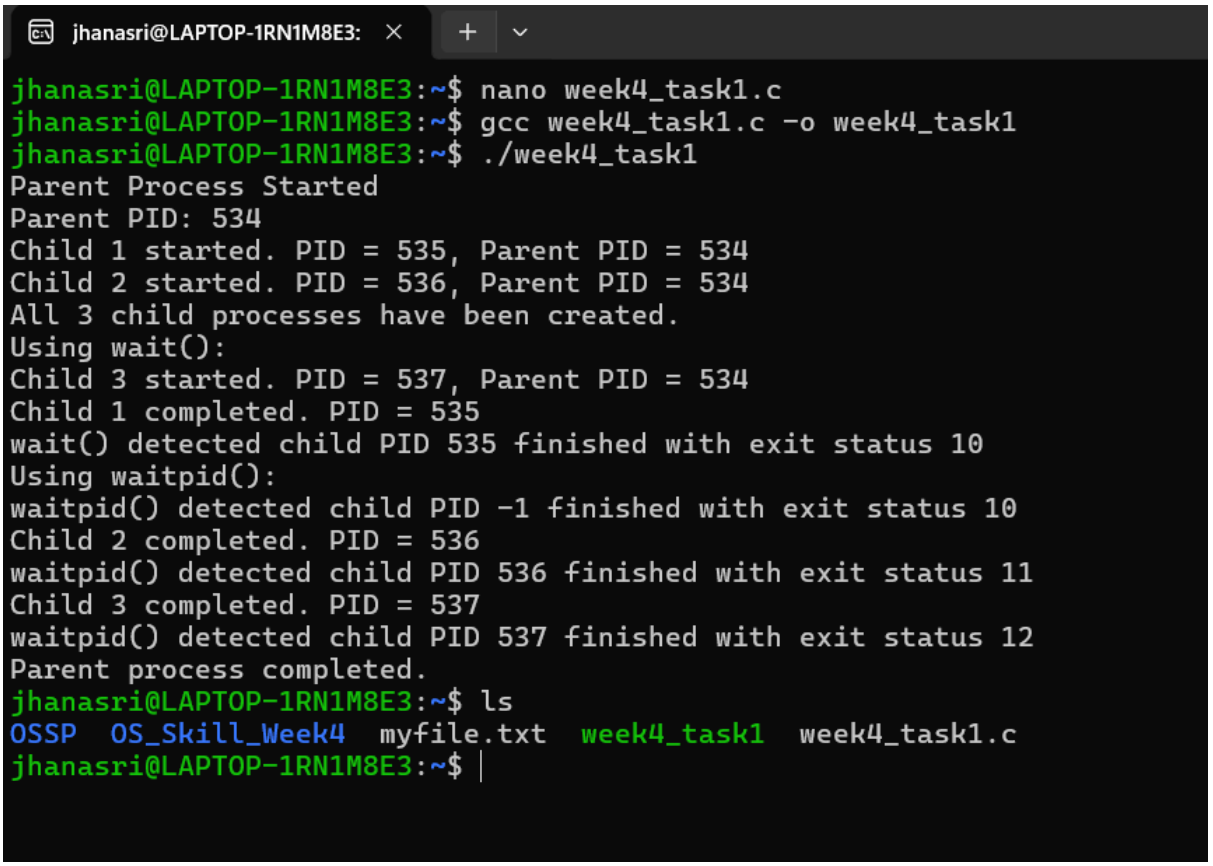
Task 3:

Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.



Step 1: Open Ubuntu Terminal

Press:

Ctrl + Alt + T

Step 2: Create a C file

Type:

nano week4_task1.c

Step 3: Enter the C program

Paste your Task 1 C program into the editor.

If you need the complete program, use:

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    pid_t child[3];
    int status;
    int i;

    printf("Parent Process Started\n");
    printf("Parent PID: %d\n", getpid());

    for (i = 0; i < 3; i++) {
        child[i] = fork();

        if (child[i] < 0) {
            perror("fork failed");
            exit(1);
        }

        if (child[i] == 0) {
            printf("Child %d started. PID = %d, Parent PID = %d\n",
                   i + 1, getpid(), getppid());

            sleep((i + 1) * 2);

            printf("Child %d completed. PID = %d\n", i + 1, getpid());

            exit(i + 10);
```

```
        }
    }

    printf("All 3 child processes have been created.\n");

    printf("Using wait():\n");

    pid_t pid = wait(&status);

    if (WIFEXITED(status)) {
        printf("wait() detected child PID %d finished with exit status %d\n",
               pid, WEXITSTATUS(status));
    }

    printf("Using waitpid():\n");

    for (i = 0; i < 3; i++) {
        pid = waitpid(child[i], &status, 0);

        if (WIFEXITED(status)) {
            printf("waitpid() detected child PID %d finished with exit status %d\n",
                   pid, WEXITSTATUS(status));
        }
    }

    printf("Parent process completed.\n");

    return 0;
}
```

Step 4: Save the file

In nano:

Ctrl + O

Press **Enter**.

Then:

Ctrl + X

Step 5: Compile the program

Type:

gcc week4_task1.c -o week4_task1

If there is **no error**, compilation is successful.

Step 6: Run the program

Type:

./week4_task1

Step 7: Observe the output

You should see something similar to:

Parent Process Started

Parent PID: 4521

Child 1 started. PID = 4522, Parent PID = 4521

Child 2 started. PID = 4523, Parent PID = 4521

Child 3 started. PID = 4524, Parent PID = 4521

All 3 child processes have been created.

Using wait():

Child 1 completed. PID = 4522

wait() detected child PID 4522 finished with exit status 10

Using waitpid():

Child 2 completed. PID = 4523

Child 3 completed. PID = 4524

waitpid() detected child PID 4523 finished with exit status 11

waitpid() detected child PID 4524 finished with exit status 12

Parent process completed.

Your practical specifically demonstrates creating **three children**, then using `wait()` and `waitpid()` to synchronize their completion.

Step 8: If you want to check the file

ls

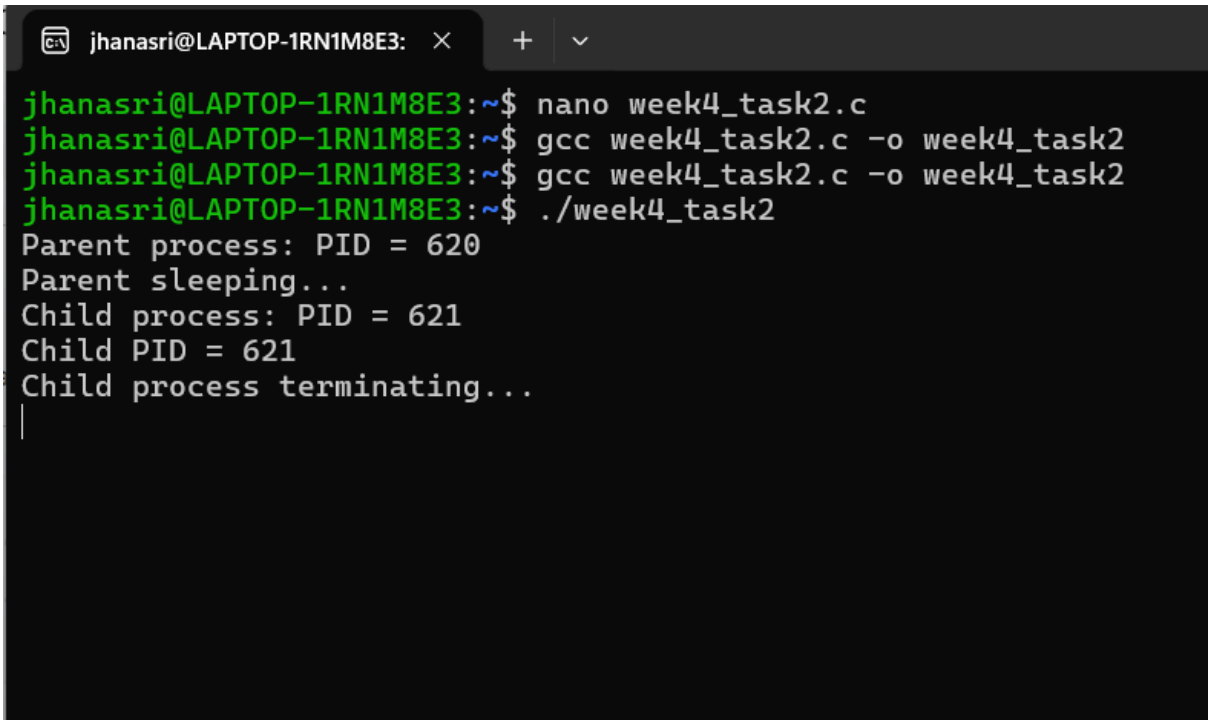
You should see:

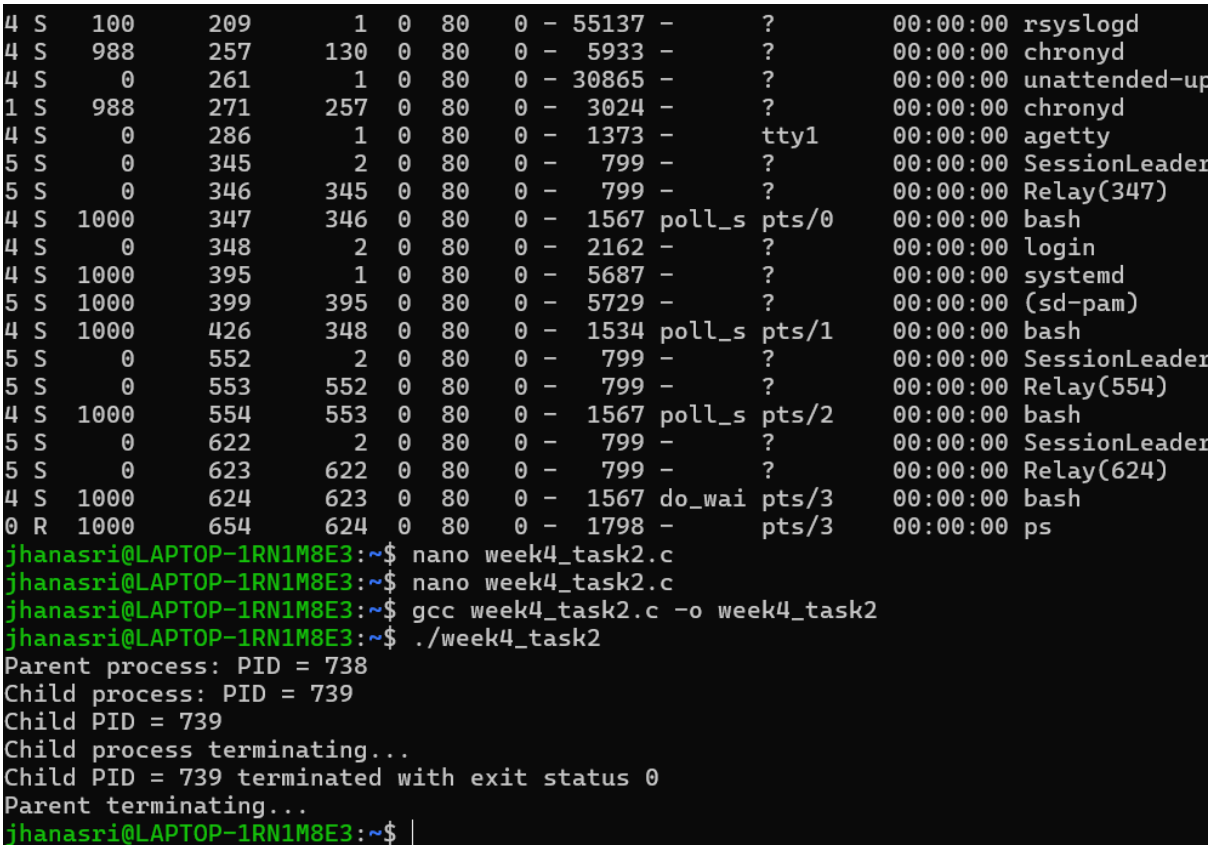
week4_task1.c

Week4_task1

Task 4:

Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.





Step 1: Open Ubuntu Terminal

Press:

Ctrl + Alt + T

Step 2: Create the C file

nano week4_task2.c

Step 3: Enter the Zombie Program

```
#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

#include <sys/types.h>


int main() {

    pid_t pid;


    pid = fork();
```

```
    if (pid < 0) {
        perror("fork failed");
        exit(1);
    }

    if (pid == 0) {
        printf("Child process: PID = %d\n", getpid());
        printf("Child PID = %d\n", getpid());

        printf("Child process terminating...\n");
        exit(0);
    }
    else {
        printf("Parent process: PID = %d\n", getpid());
        printf("Parent sleeping...\n");

        sleep(20);

        printf("Parent terminating...\n");
    }

    return 0;
}
```

This follows the first part of your practical: the child terminates while the parent remains alive/sleeping, creating the opportunity to observe the zombie.

Step 4: Save the program

Press:

Ctrl + O

Press **Enter**.

Then press:

Ctrl + X

Step 5: Compile the program

gcc week4_task2.c -o week4_task2

If no error appears, compilation is successful.

Step 6: Run the program

./week4_task2

You will get something like:

Parent process: PID = 158

Parent sleeping...

Child process: PID = 159

Child PID = 159

Child process terminating...

Step 7: Quickly open another terminal

While the **parent is sleeping**, open another terminal:

Ctrl + Alt + T

Step 8: Check the process table

Run:

ps -el

Step 9: Find your program

Look for:

week4_task2

The child should appear with **Z** in the status column.

Example from your practical:

0 Z 1000 159 158 ... week4_T2

Here, **Z = Zombie process**.

Step 10: Go back to the first terminal

Wait for:

Parent terminating...

The parent then finishes.

Now Eliminate the Zombie

Step 11: Open the C file again

nano week4_task2.c

Step 12: Add `waitpid()`

First add this header:

#include <sys/wait.h>

Then, inside the **parent section**, add:

int status;

waitpid(pid, &status, 0);

So the parent part becomes:

```
else {
    printf("Parent process: PID = %d\n", getpid());


    waitpid(pid, &status, 0);


    printf("Parent terminating...\n");
}
```

Step 13: Save

Ctrl + O

Press **Enter**.

Then:

Ctrl + X

Step 14: Compile again

gcc week4_task2.c -o week4_task2

Step 15: Run again

./week4_task2

Step 16: Check the process table

While the program is running, open another terminal:

ps -el

Step 17: Check for `Z`

Previously, you saw:

Z

Now the parent uses `waitpid()` to collect the child's termination status, so the child should **not remain as a zombie**.

Step 18: Finish

The parent prints:

Parent terminating...

and the program ends.